Does Gravity See Slow Mantle Regions as Heavy or Light?

A Global-Scale Comparison of Mantle Models with Observed Gravity

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Abstract

A central issue in the current LLSVP debate is whether slow seismic provinces should be treated as dense structures. That assumption is often used to argue that they should remain near the equator rather than occupy a non-equatorial degree 2 configuration. I tested the sign of that assumption directly against Earth's large scale static gravity field. At the global degree 2, or quadrupole, scale, gravity fits better when slow seismic anomalies are treated as relatively lighter. Using a reference compatible central gravity residual, seven of the eight full-mantle model fields favored the globally reversed sign. Six of the seven velocity based models therefore favored the slow-as-light mapping. The seventh reversed result came from a native density related model. Across 10,000 realizations using the documented formal gravity coefficient covariance, every model retained its original classification: seven remained reversed and one remained adopted. This result does not show that LLSVP material is uniformly buoyant, and it does not establish that a non equatorial degree 2 configuration would be dynamically stable. The conclusion applies to this large-scale static gravity comparison: at degree 2, the observed gravity field more often favored the sign in which slow seismic anomalies behave as relatively lighter mass anomalies. It does not directly determine LLSVP density, composition, rheology, or behavior during mantle displacement.

Keywords: seismic tomography; gravity field; degree 2; sign convention; mantle density; covariance; principal axes

1 Background

Seismic tomography mainly shows variations in seismic wave speed. Converting those velocity patterns into density requires additional assumptions, joint inversions, mineral physics relationships, or an independently supplied density field [1–3]. A slow anomaly can be interpreted differently depending on how temperature, composition, phase, attenuation, anisotropy, and the reference state are handled. The velocity map itself does not tell us whether the anomaly is heavy or light.

Once a tomographic model is converted into a gravity or inertia tensor, that sign choice becomes part of the calculation. It controls whether a slow anomaly contributes as a positive or negative mass anomaly, and whether the same eigendirection is labeled as a minimum- or maximum-moment direction. The direction can stay fixed while its physical role changes.

Instead of choosing the sign by convention, I compared both possibilities with the observed gravity field. I gave gravity a vote.

This is a focused test. It compares full-mantle degree 2 model vectors with a declared reference-compatible static-gravity residual. The result applies to the large scale coefficient geometry under the stated residual and covariance assumptions. It is not a direct measurement of uniform LLSVP density and it does not replace thermodynamic or mineral physics analysis.

2 Model setup

2.1 Observed degree 2 gravity residual

The coefficient vector is ordered as

$$\boldsymbol{d} = [C_{20}, C_{21}, S_{21}, C_{22}, S_{22}]^T . \tag{1}$$

The primary branch, `R_compatible_central`, was derived from GOCO2025s by subtracting the declared IERS reference figure and the validated compatible mean ocean, grounded Antarctic ice, and grounded Greenland ice coefficient vectors. The coefficients are fully normalized, zero tide, and referenced to a radius of 6.378×10^6 m. The observed gravity coefficients were taken from GOCO2025s [5]. The axisymmetric reference figure follows IERS Conventions (2010) [4].

This branch is explicitly partial. It does not include a reference compatible LITHO1.0 subtraction, topographic or crustal compensation closure, atmosphere, hydrology, mountain glaciers, glacial isostatic adjustment, or a complete covariance for the correction vectors. Table 1 lists the central residual and the formal coefficient uncertainties used in the fit.

2.2 Full-mantle model fields

Eight full-mantle fields were evaluated: S40RTS, TX2011, SEISGLOB2, SPani, SEMUCB-WM1, CSEM2-Globe-RHO, CSEM2-Globe-VSV, and UNICA25-absolute-Vp. These span global S-wave tomography, anisotropic S-wave tomography, a native density-related field, a vertical shear

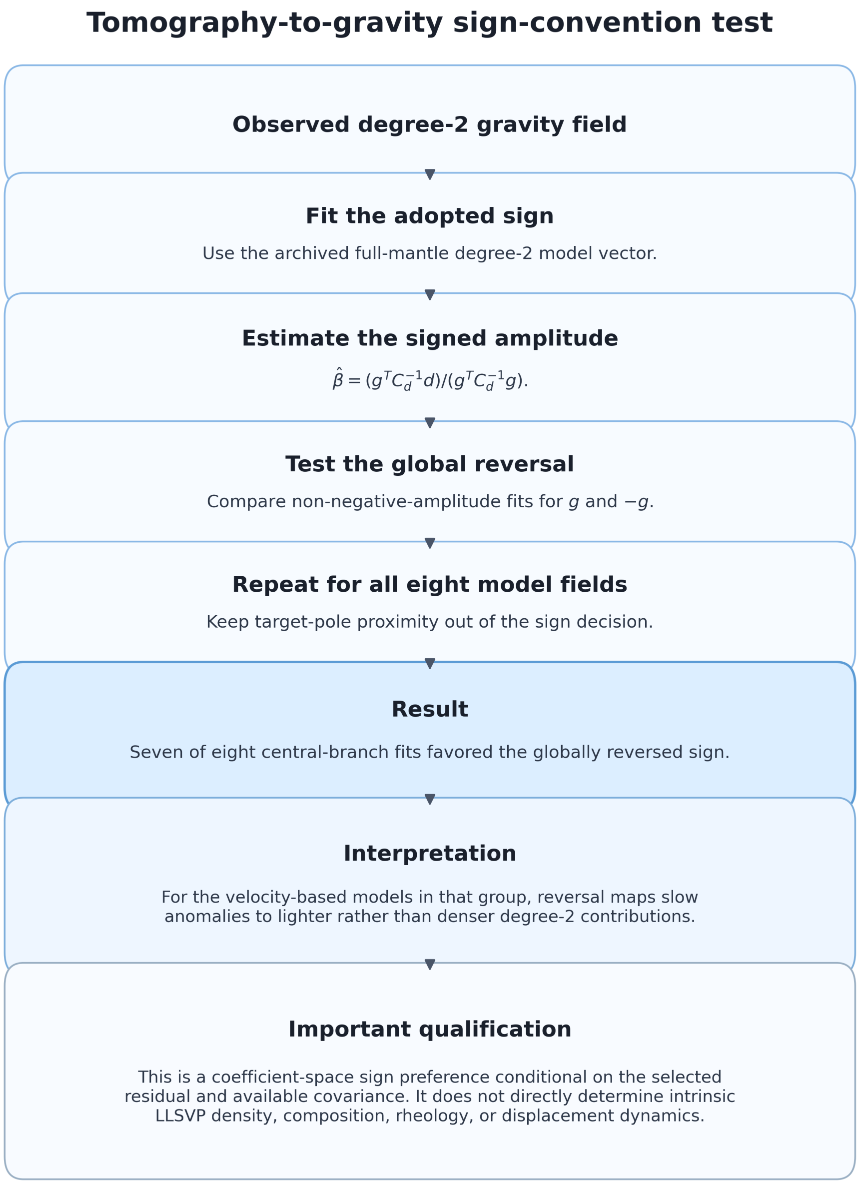

Figure 1: Workflow for the sign convention test. The sign decision is made entirely in degree 2 coefficient space. Target pole proximity is excluded from the fit, so the test does not select a sign because it moves a preferred axis toward a desired location. The final box states the central limitation: this is a conditional sign preference rather than a direct density or dynamics measurement.

Table 1: Primary residual coefficients and formal standard deviations. The coefficient uncertainties come from the archived reference-figure gravity product. Uncertainty in the compatible correction vectors is not included.

Coefficient	Central residual	Formal σ
C_{20}	6.055629264e-06	2.000019572e-11
C_{21}	5.795538126e-06	8.775592797e-14
S_{21}	4.910665758e-06	9.787141450e-14
C_{22}	-3.803033039e-06	1.359212511e-13
S_{22}	-3.376437631e-06	1.298328942e-13

velocity field, and a depth-referenced P-wave anomaly product [6–12].

Seven entries are velocity based. Their adopted conversion used a multiplier of -1, corresponding to the project's original slow-as-dense proxy. CSEM2-Globe-RHO is different: it is a native density-related field whose adopted sign was retained.

For each model m, the archived degree-2 prediction is

$$\boldsymbol{g}_m = [g_{20}, g_{21}, g_{S21}, g_{22}, g_{S22}]^T . \tag{2}$$

The model vectors are shape vectors. Their normalization changes the fitted amplitude but cannot change the sign preference, because positive rescaling of $\boldsymbol{g}_m$ rescales $\widehat{\beta}$ inversely while preserving $\widehat{\beta}\boldsymbol{g}_m$.

2.3 Signed coefficient-space fit

The observed residual is fit as

$$\boldsymbol{d} \approx \beta \boldsymbol{g}_m. \tag{3}$$

With $\boldsymbol{W} = \boldsymbol{C}_d^{-1}$, the unregularized weighted least-squares estimator is

$$\widehat{\beta}_m = \frac{\boldsymbol{g}_m^T \boldsymbol{W} \boldsymbol{d}}{\boldsymbol{g}_m^T \boldsymbol{W} \boldsymbol{g}_m}, \tag{4}$$

and its formal standard error is

$$\sigma_{\beta,m} = \left(\boldsymbol{g}_m^T \boldsymbol{W} \boldsymbol{g}_m\right)^{-1/2} . \tag{5}$$

The sign rule is direct:

$$\widehat{\beta}_m > 0 \quad \Rightarrow \quad \text{adopted sign}, \tag{6}$$

$$\widehat{\beta}_m < 0 \quad \Rightarrow \quad \text{globally reversed sign}. \tag{7}$$

A fit is treated as indeterminate when $|\widehat{\beta}| \leq 2\sigma_\beta$ or the formal 95% interval crosses zero. No regularization term was used in this branch.

The two constrained hypotheses use a non-negative amplitude under each sign:

$$a_{\text{adopted}} = \max(0, \widehat{\beta}), \tag{8}$$

$$a_{\text{reversed}} = \max(0, -\widehat{\beta}). \tag{9}$$

Their weighted residuals are

$$\chi^2_{\text{adopted}} = (\boldsymbol{d} - a_{\text{adopted}}\boldsymbol{g})^T \boldsymbol{W} (\boldsymbol{d} - a_{\text{adopted}}\boldsymbol{g}), \tag{10}$$

$$\chi^2_{\text{reversed}} = (\boldsymbol{d} + a_{\text{reversed}}\boldsymbol{g})^T \boldsymbol{W} (\boldsymbol{d} + a_{\text{reversed}}\boldsymbol{g}). \tag{11}$$

Thus, $\Delta\chi^2 = \chi^2_{\text{adopted}} - \chi^2_{\text{reversed}} > 0$ favors reversal. This constrained comparison avoids the trivial identity between an unrestricted fit to $\boldsymbol{g}$ and an unrestricted fit to $-\boldsymbol{g}$.

2.4 Covariance and Monte Carlo propagation

The available covariance is diagonal:

$$\boldsymbol{C}_d = \operatorname{diag}\left(\sigma^2_{C20}, \sigma^2_{C21}, \sigma^2_{S21}, \sigma^2_{C22}, \sigma^2_{S22}\right) . \tag{12}$$

Ten thousand realizations were drawn:

$$\boldsymbol{d}^{(r)} \sim \mathcal{N}(\boldsymbol{d}, \boldsymbol{C}_d), \tag{13}$$

with the compatible correction vectors held fixed. The resulting intervals therefore quantify only the documented formal coefficient uncertainty. They are not a complete uncertainty for the mantle residual.

2.5 What changes when the sign is reversed

Let the trace free degree 2 tensor be $\boldsymbol{Q}$, with eigenvectors $\boldsymbol{v}_i$ and eigenvalues λ_i:

$$\boldsymbol{Q}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i. \tag{14}$$

Reversing the sign of the tensor gives

$$(-\boldsymbol{Q})\boldsymbol{v}_i = (-\lambda_i)\boldsymbol{v}_i. \tag{15}$$

The axis directions do not move. The eigenvalues simply change sign, which reverses their order. The direction labeled as the minimum under one sign becomes the maximum under the opposite sign, and the maximum becomes the minimum. The intermediate direction stays intermediate. Figure 2 shows this role swap.

2.6 Validation workflow

The public script ran successfully using the ten frozen files in the `data/` and `validation/` folders. It reproduced the saved fit results, formal uncertainties, constrained sign comparisons, and Monte Carlo summaries. It also recorded a SHA-256 checksum for each public input so the exact files used in the calculation can be verified.

The repository contains everything needed to rerun the sign test from the frozen degree 2 inputs: the gravity-residual coefficients, the eight model vectors, the documented coefficient uncertainties, the fitting rules, the Monte Carlo settings, and the standalone Python script.

Reproduction begins with the finished degree 2 vectors. The earlier steps used to create those vectors from the

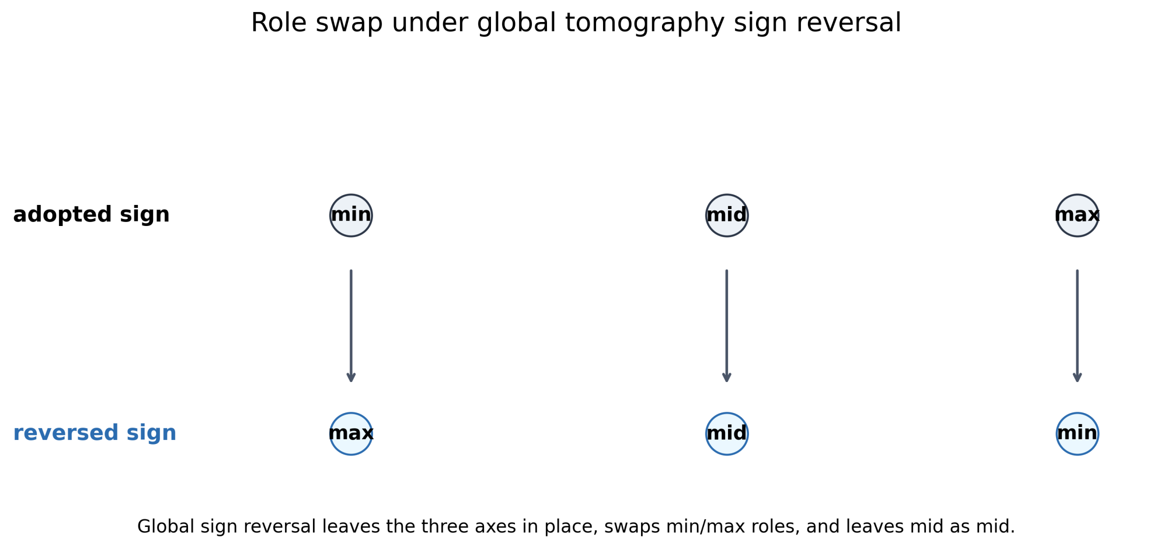

Figure 2: Global sign reversal changes eigenvalue ordering without rotating the eigendirections. A location identified by the minimum eigenvalue under the adopted sign becomes the maximum-eigenvalue direction under reversal; the intermediate direction remains intermediate. This role swap is exact apart from floating-point eigensolver precision.

original three dimensional tomography models are outside the public package. Rebuilding the analysis from the raw source data would also require the original tomography grids, gravity products, shallow correction data, model-reading code, preprocessing notebooks, software environment, and complete processing history.

3 Results

3.1 Deterministic sign fits

Figure 3 shows the fitted signed amplitudes. Seven model fields lie on the negative side of zero. CSEM2-Globe-VSV is the single positive central-branch result.

The central result is:

> **For seven of the eight full-mantle model fields, the global scale static gravity comparison favored the globally reversed sign. Among the seven velocity based fields, six favored the mapping in which slow seismic anomalies contribute as relatively lighter rather than denser mass anomalies.**

One of the seven reversed results comes from CSEM2-Globe-RHO, which is based on a density related field rather than seismic velocity. Among the seven velocity based models, six favor the slow-as-light mapping, while CSEM2-Globe-VSV favors the usual slow-as-dense mapping.

So, the seven of eight result applies to all model fields together. For the velocity-based models alone, the result is six of seven.

At degree 2, gravity more often matched the reversed tomography sign.

Table 2 gives the model-level results. All seven negative central-branch values are many formal standard errors below zero under the available covariance, while the CSEM2-Globe-VSV value is many formal standard errors above zero.

3.2 Constrained residual comparison

The second comparison gave the same result. For each model, I fit both the adopted sign and the reversed sign while requiring the fitted amplitude to remain non-negative. The reversed sign produced the smaller weighted residual for seven of the eight model fields. CSEM2-Globe-VSV was the only model that fit better with the adopted sign.

3.3 Monte Carlo sign stability

Figure 5 shows the results from 10,000 Monte Carlo realizations. Seven model fields produced $\beta < 0$ every time, while CSEM2-Globe-VSV produced $\beta > 0$ every time.

This strong split comes from the very small formal uncertainties assigned to the gravity coefficients compared with the size of the fitted values. These runs test only the uncertainties that were included in the calculation, so they should not be treated as complete physical confidence.

The sign stayed the same under every gravity-coefficient perturbation included in the Monte Carlo test. This result applies only to the uncertainties modeled here. It does not mean the physical interpretation is known with 100% confidence.

3.4 Residual and coefficient sensitivity

The overall count stayed the same across all three residual choices: seven models favored the reversed sign and one favored the adopted sign. The one exception changed

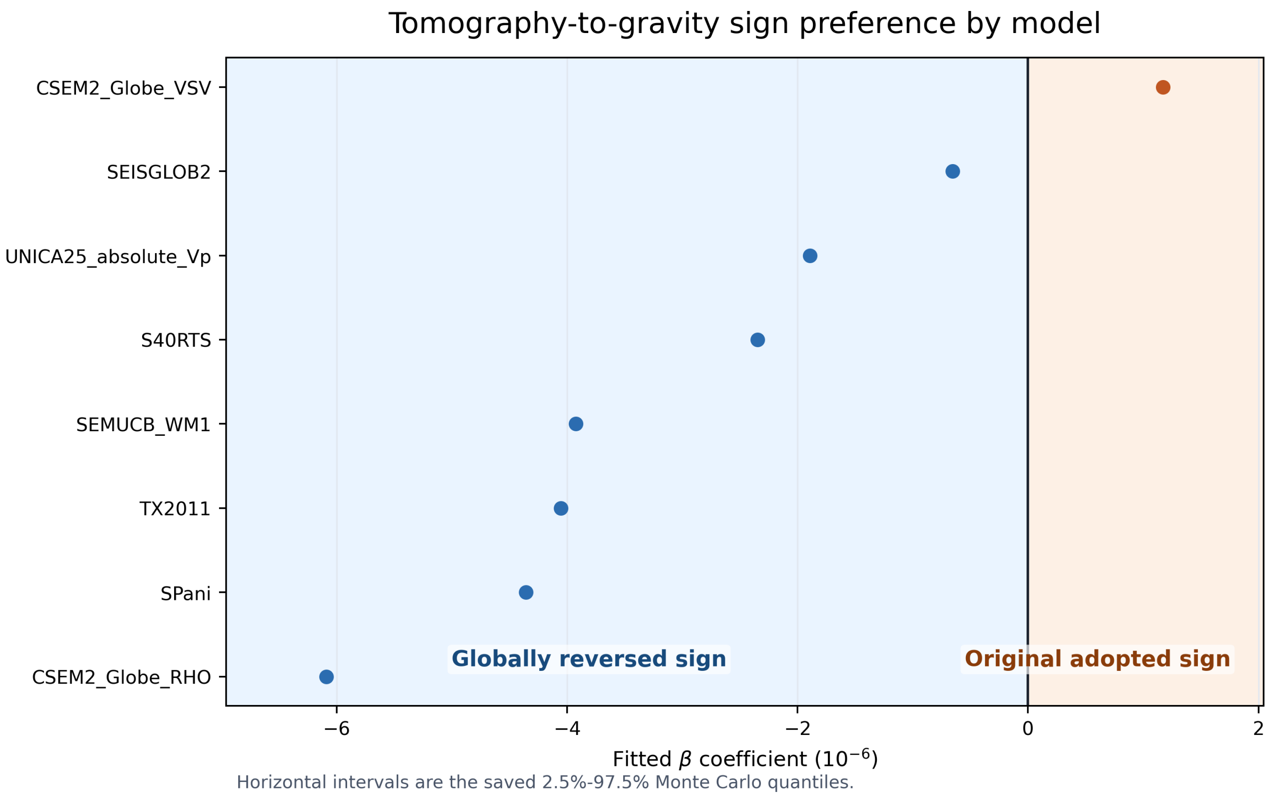

Figure 3: Signed β estimates for the primary reference-compatible residual. Negative values support a global reversal of the adopted model sign; positive values support the adopted sign. The horizontal intervals are the saved 2.5–97.5% Monte Carlo quantiles and are visually small because they propagate only the documented formal gravity-coefficient covariance.

Table 2: Model-by-model sign results for the primary residual. β and the Monte Carlo interval are reported in units of 10^{-6}. "Fraction $\beta < 0$" refers only to the documented formal covariance with correction vectors held fixed.

Model	Input quantity	$\widehat{\beta}$	MC 95% interval	Fraction $\beta < 0$	Preferred	Model-level interpretation
S40RTS	V_S perturbation	-2.347340	[-2.347340, -2.347339]	100.0%	Reversed	slow-as-light
TX2011	V_S perturbation	-4.054710	[-4.054711, -4.054710]	100.0%	Reversed	slow-as-light
SEISGLOB2	V_S perturbation	-0.656103	[-0.656103, -0.656102]	100.0%	Reversed	slow-as-light
SPani	anisotropic V_S perturbation	-4.357798	[-4.357798, -4.357798]	100.0%	Reversed	slow-as-light
SEMUCB_WM1	V_S perturbation	-3.923406	[-3.923406, -3.923406]	100.0%	Reversed	slow-as-light
CSEM2_Globe_RHO	native density-related field	-6.089010	[-6.089010, -6.089009]	100.0%	Reversed	reversed native density-field sign
CSEM2_Globe_VSV	V_{SV}-related field	1.172924	[1.172924, 1.172925]	0.0%	Adopted	slow-as-dense
UNICA25_absolute_Vp	depth-mean V_P anomaly	-1.892465	[-1.892466, -1.892465]	100.0%	Reversed	slow-as-light

Table 3: Constrained weighted residual norms. Values are shown in units of 10^7. The percent change is $(r_{\text{adopted}} - r_{\text{reversed}})/r_{\text{adopted}}$; positive values favor reversal.

Model	Adopted residual	Reversed residual	Change	Favored sign
S40RTS	9.1314	8.9471	+2.019%	Reversed
TX2011	9.1314	8.6835	+4.905%	Reversed
SEISGLOB2	9.1314	9.1167	+0.161%	Reversed
SPani	9.1314	8.5870	+5.962%	Reversed
SEMUCB_WM1	9.1314	8.6643	+5.115%	Reversed
CSEM2_Globe_RHO	9.1314	8.1675	+10.556%	Reversed
CSEM2_Globe_VSV	9.0945	9.1314	-0.405%	Adopted
UNICA25_absolute_Vp	9.1314	9.0400	+1.001%	Reversed

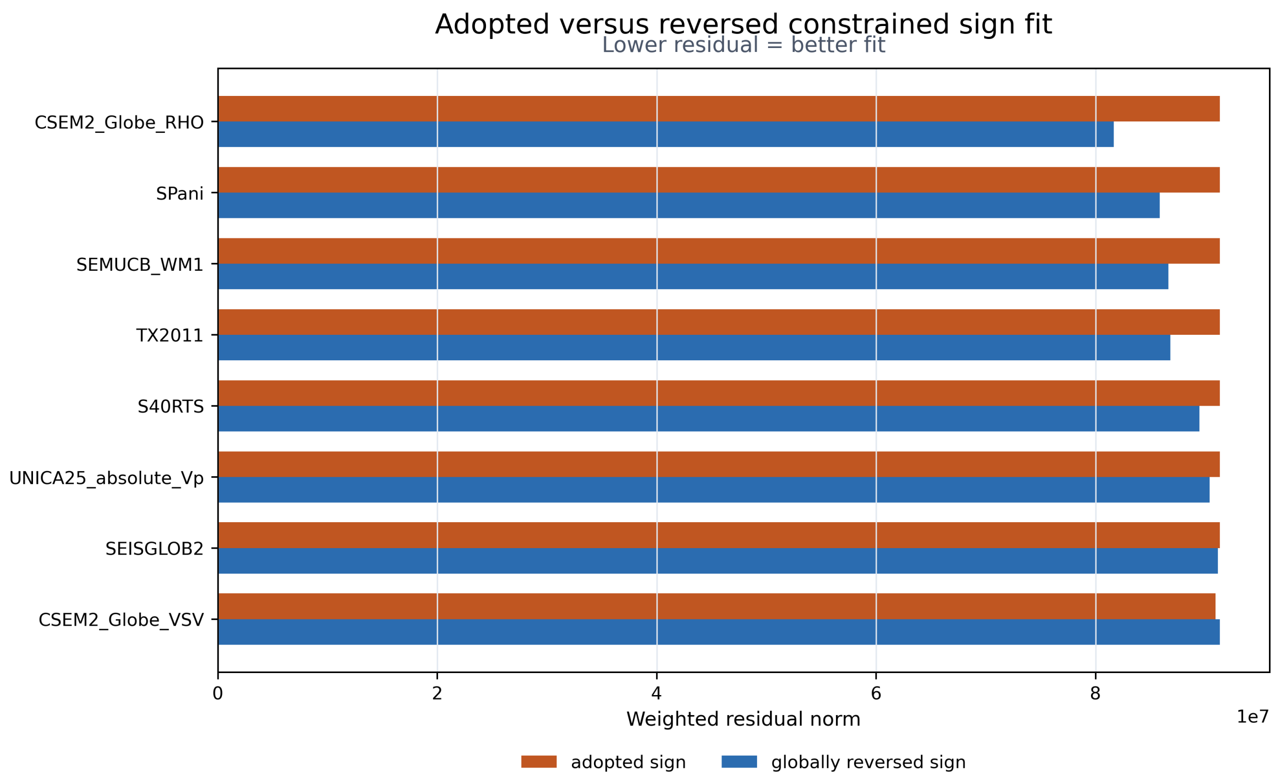

Figure 4: Comparison of the weighted residuals for the adopted and reversed signs. A smaller residual means a better fit to the observed gravity field under the stated covariance assumptions. Seven model fields fit better with the reversed sign. CSEM2-Globe-RHO shows the largest improvement, while SEISGLOB2 shows the smallest improvement among the reversed cases. This figure compares fit quality within this test and does not represent a complete physical probability for either interpretation.

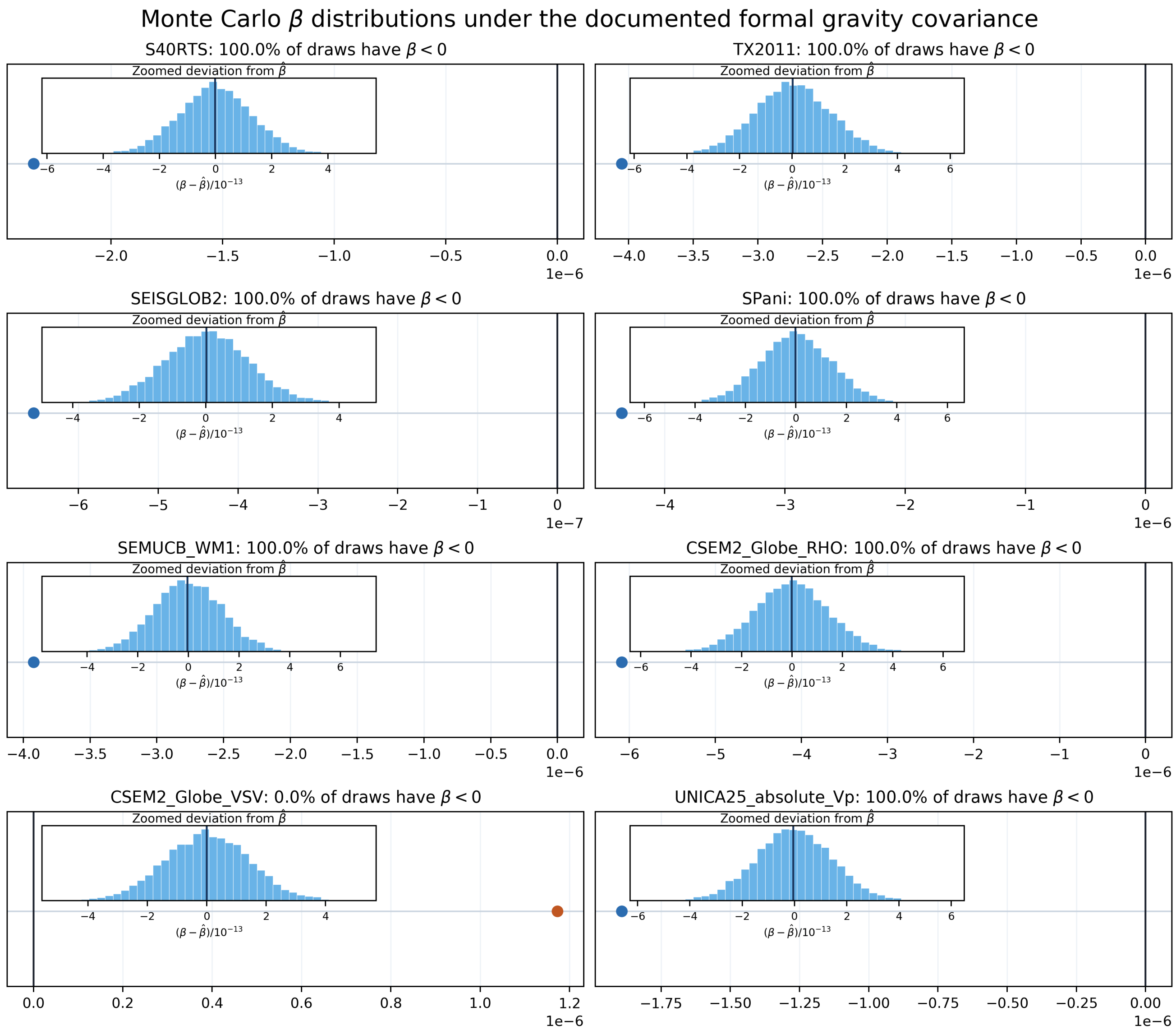

Figure 5: Results from 10,000 Monte Carlo realizations using the documented diagonal covariance. Each main panel shows the fitted value and zero on the same scale, while the inset zooms in on the distribution around $\widehat{\beta}$. These distributions include only the formal gravity coefficient uncertainty. They do not include uncertainty in the shallow corrections, gravity reference closure, tomography model structure, or unavailable coefficient correlations.

Table 4: Sign-count sensitivity.

Case	Rev.	Adopt.	Indet.
Reference residual	7	1	0
Central residual	7	1	0
LITHO1 anomaly branch	7	1	0
Omit C_{20}	7	1	0
Omit C_{21}	8	0	0
Omit S_{21}	8	0	0
Omit C_{22}	6	2	0
Omit S_{22}	4	4	0

depending on the residual. UNICA25-absolute-Vp favored the adopted sign for the reference residual, while CSEM2-Globe-VSV favored the adopted sign for the central and anomaly-sensitivity residuals.

I also repeated the test after removing one gravity coefficient at a time. Removing C_{21} or S_{21} made all eight models favor the reversed sign. Removing C_{22} gave six reversed and two adopted. Removing S_{22} changed the result the most, producing an even split of four reversed and four adopted.

Figure 6 and Table 4 show how the result changes under each test.

Figure 6: How the sign count changes with different residual choices and after removing one gravity coefficient at a time. The seven-to-one result stays the same across all three residual branches. Removing S_{22} has the largest effect, changing the result to four reversed and four adopted.

4 Interpretation

Observation. Seven of the eight model fields in the central branch gave $\widehat{\beta} < 0$. The reversed sign also produced the smaller residual for those same seven models. The overall count stayed at seven reversed and one adopted across all three residual branches.

Interpretation. At degree 2, the observed gravity residual usually matched the reversed model pattern better than the adopted one. For the velocity based models, this means the slow regions fit better when they were treated as relatively lighter rather than denser.

Physical inference. Within this large scale comparison, the slow-as-light mapping was favored more often than the usual slow-as-dense mapping. This does not mean every part of the LLSVPs has the same density or physical state. Local density can differ from the global degree-2 pattern, and one of the eight inputs is a native density-related field.

The result applies to this degree-2 coefficient comparison and depends on the chosen gravity residual and covariance model.

It is not a direct measurement of the true density of LLSVP material. It comes from comparing full-mantle model patterns with the observed gravity field.

The uncertainty ranges include the formal gravity-coefficient uncertainty used in the calculation. They do not include the full uncertainty in the shallow-mantle corrections, so they should not be read as a complete uncertainty budget.

This analysis does not determine:

- intrinsic density,
- composition,
- mantle rheology, or
- displacement dynamics.

The result also does not prove that every slow anomaly is buoyant. It reports the sign of the best-fitting degree-2 projection. Higher-degree structure, radial variation, compensation, chemical heterogeneity, and local departures from the fitted global relation remain outside the test.

5 Discussion

This analysis tests the density sign assumption often used to argue that LLSVPs should remain near the equator. It does not determine which orientation would be mechanically stable.

5.1 What changes

The adopted tomography-to-density sign should not be assumed to be the better choice at degree 2. The static gravity residual favored the reversed sign for seven of the eight model fields in the central branch. The same seven-to-one count also appeared in the two alternative residual branches, so the result is not limited to a single residual choice.

Reversing the sign does not move the axes. It only reverses their roles. The maximum-moment direction becomes the minimum-moment direction, the minimum becomes the maximum, and the intermediate direction stays intermediate. The geometry stays the same; the labels switch.

5.2 What remains conditional

The Monte Carlo test covers only part of the uncertainty. The correction vectors were held fixed because their covariance was unavailable, and the central residual includes only part of the possible shallow mass corrections. The result also depends strongly on S_{22}: removing that coefficient changes the count from seven reversed and one adopted to an even four-to-four split.

So the sign result should remain conditional, even though each fitted β is far from zero under the formal uncertainties that were included.

The sensitivity tests also show two different kinds of stability. The overall seven-to-one count stays the same across all three residual branches, while the identity of the single adopted model changes between the reference residual and the two corrected residuals.

5.3 The remaining dynamical question

The remaining question is dynamical: whether buoyant lowermost structures would move coherently with the mantle shell, deform relative to the core-mantle boundary, or re-equilibrate after displacement.

This paper does not answer that question. A coefficient-space sign preference does not prescribe mechanical coupling, relaxation time, viscosity, phase behavior, or survival of lowermost-mantle structures during large reorientation. Those questions require a separate dynamical model with explicit forces, constitutive behavior, boundary conditions, and timescales.

This study shows which sign the observed degree-2 gravity field favors more often.

6 Conclusion

I compared the adopted and reversed tomography-to-gravity signs with a declared degree 2 static-gravity residual. In the primary branch, seven of the eight full-mantle model fields favored the reversed sign. The constrained residual comparison gave the same result. Under the formal coefficient uncertainty used here, those same seven models kept $\beta < 0$ in all 10,000 Monte Carlo realizations.

Among the seven velocity based models, six favored the slow-as-light mapping at degree 2. The seventh reversed result came from a native density related field. The clearest summary is therefore seven of eight model fields favoring the reversed sign, with six of seven velocity based models favoring slow-as-light.

The result depends on the chosen reference compatible residual, the diagonal covariance, the fixed correction vectors, and the limited uncertainty available for the shallow corrections. Removing S_{22} changes the result from seven reversed and one adopted to four reversed and four adopted. The identity of the single adopted model also changes when the uncorrected reference residual is used.

Future work should improve the uncertainty treatment for shallow loads, test other gravity products and reference states, include uncertainty in the tomography models, examine how the sign changes with depth, and then study the separate question of how lowermost-mantle structures might move, deform, or settle into a new position.

Disclosure and availability

This is an independent geophysical modeling case study. The public repository includes the paper, the frozen degree 2 input files, the saved validation results, and the standalone code needed to rerun the sign test.

The private working notebooks, raw three-dimensional tomography models, larger intermediate datasets, and earlier preprocessing code are not included. Reproduction begins with the final degree 2 coefficient vectors used in the paper.

The public script checks the reproduced results against the saved validation files and records a checksum for each public input.

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